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RECIPROCALS OF SUBSUM POLYNOMIALS

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Abstract. We introduce the subsum polynomial of a partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ defined by $\mathrm{sp}(\lambda, x) = \prod_{i=1}^{k}(1 + x^{\lambda_i})$. We study the sum of reciprocals of $\mathrm{sp}(\lambda, x)$ over all partitions of n. We prove arithmetic properties of related polynomials and offer connections to other combinatorial objects.

1. Introduction

An integer partition λ of n is a nonincreasing sequence $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ of positive integers that add up to n. We write $\lambda \vdash n$ to mean that λ is a partition of n. The numbers λ_i in the sequence are called the parts of λ. We identify a partition with its multiset of parts. We denote by $P(n)$ the number of partitions of n and set $P(0) := 1$. We occasionally use the frequency notation of partitions. Exponents of parts represent their multiplicity. For example, $(3^2, 2^4, 1^2)$ is the partition $(3, 3, 2, 2, 2, 2, 1, 1)$. For more on the theory of partitions, we refer the reader to [1].

Given a partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$, we define the polynomial

$$\mathrm{sp}(\lambda, x) := \prod_{i=1}^{k}(1 + x^{\lambda_i}).$$

In analogy with the generating function for the number of partitions with distinct parts, we can view the polynomial $\mathrm{sp}(\lambda, x)$ as the generating function for the number of partitions whose parts form a submultiset of λ. This inspired the name *subsum polynomial* for $\mathrm{sp}(\lambda, x)$. This construction is also reminiscent of Stanton's polynomial G_λ, the generating function of all partitions that lie inside of λ (see [7]). Clearly, if λ is a partition of n, then $\deg(\mathrm{sp}(\lambda, x)) = n$.

If $m_\lambda(i)$ denotes the number of times i occurs as a part in λ, then

$$\mathrm{sp}(\lambda, x) = \prod_{i=1}^{n}(1 + x^i)^{m_\lambda(i)}.$$

Numerical experimentation led us to define and study the rational function

$$\mathrm{sr}(n, x) := \sum_{\lambda \vdash n} \frac{1}{\mathrm{sp}(\lambda, x)}.$$

For any partition $\lambda \vdash n$, let

$$h_\lambda(x) := \prod_{i \geq 1}(1 + x^i)^{\lfloor n/i \rfloor - m_\lambda(i)}.$$

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We define

$$\mathrm{num}^*(n,x) := \sum_{\lambda \vdash n} h_\lambda(x)$$

and

$$\mathrm{den}^*(n,x) := \prod_{i \geq 1} (1+x^i)^{\lfloor n/i \rfloor}.$$

Then,

$$\mathrm{sr}(n,x) = \frac{\mathrm{num}^*(n,x)}{\mathrm{den}^*(n,x)}.$$

We can also write

$$\mathrm{num}^*(n,x) = \sum_{\substack{\mu \vdash \sum_{i=1}^n \sigma_1(i) - n \\ m_\mu(i) \leq \lfloor \frac{n}{i} \rfloor}} \mathrm{sp}(\mu, x),$$

where $\sigma_1(i)$ denotes the sum of divisors of i.

The degree of $\mathrm{den}^*(n,x)$ is $\sum_{i=1}^n i\lfloor n/i \rfloor = \sum_{i=1}^n \sigma_1(i)$. This is sequence A024916 in OEIS [5]. We have

$$\deg(h_\lambda(x)) = \sum_{i=1}^n \sigma_1(i) - \sum_{i=1}^n i m_\lambda(i) = \sum_{i=1}^n \sigma_1(i) - n = \deg(\mathrm{num}^*(n,x)).$$

Thus

$$\deg(\mathrm{den}^*(n,x)) - \deg(\mathrm{num}^*(n,x)) = n.$$

Next, we define

$$G(n,x) := \gcd\{h_\lambda(x) \mid \lambda \vdash n\},$$

and let

$$\mathrm{num}(n,x) := \frac{\mathrm{num}^*(n,x)}{G(n,x)} \quad \text{and} \quad \mathrm{den}(n,x) := \frac{\mathrm{den}^*(n,x)}{G(n,x)}.$$

In Section 3 we prove results about $\mathrm{num}(n,x)$ and $\mathrm{den}(n,x)$. Below we list some of them.

A polynomial is *palindromic* if its coefficients form a palindrome. A sequence $u_1, u_2, \ldots, u_m$ of real numbers is said to be *unimodal* if, for some j, $u_1 \leq u_2 \leq \ldots \leq u_j \geq u_{j+1} \geq \ldots \geq u_m$. A polynomial is unimodal if its sequence of coefficients is unimodal.

Proposition (3.9)**.** *Let n be a positive integer. Then,*

(a) *$G(n,x)$, $\mathrm{num}(n,x)$, and $\mathrm{den}(n,x)$ are palindromic.*
(b) *$\mathrm{den}(n,x)$ and $G(n,x)$ are unimodal.*

Theorem (3.18)**.** *For all $n \geq 1$,*

(a) $\mathrm{num}(n,-1) = n!$
(b) $|\mathrm{num}(n,\pm \mathrm{i})| = 2^{\lfloor n/2 \rfloor}\lfloor n/2 \rfloor!$
(c) $|\mathrm{num}(n,\zeta_6)| = \begin{cases} 3^{\lfloor n/3 \rfloor}\lfloor n/3 \rfloor! & n \not\equiv 2 \pmod 3 \\ 4 \cdot 3^{\lfloor n/3 \rfloor}\lfloor n/3 \rfloor! & n \equiv 2 \pmod 3, \end{cases}$
where ζ_6 is a primitive sixth root of unity.

Proposition (3.13)**.** *For all $n \geq 1$,*

(a) $\mathrm{den}(n,1) = 2^{\mathrm{val}_2((2n)!)}$,
(b) $\mathrm{num}(n,1) = 2^{\mathrm{val}_2(n!)}P(n)$.

Numerical evidence suggests the following conjecture.

Conjecture 1. *For $n \geq 1$, the polynomial $\operatorname{num}(n,x)$ is irreducible over the integers.*

We also make a somewhat weaker claim for which we have achieved some partial results:

Conjecture 2. *For $n \geq 1$,*

$$\gcd\big(\operatorname{num}(n,x), \operatorname{den}(n,x)\big) = 1\,.$$

Conjecture 1 implies Conjecture 2 (see Remark 5).

In Section 4 we prove some analogous results for binary partitions, that is, partitions whose parts are powers of two. In Section 5, we briefly consider a generalization of $\operatorname{sr}(n,x)$, namely $\sum_{\lambda\vdash n} \operatorname{sp}(\lambda,x)^m$ for $m \in \mathbb{Z}$. In Section 6, we discuss open questions for sums of reciprocals of subsum polynomials defined on restricted sets of partitions.

2. Preliminaries

In this section we establish useful properties of polynomials of the form $1+x^i$. For the convenience of the reader, we give complete proofs.

We write $\operatorname{val}_2(n)$ for the 2-valuation of the integer n. We denote by $\Phi_d(x)$ the d-th cyclotomic polynomial, which is irreducible in $\mathbb{Z}[x]$.

Remark 1. Since $x^n - 1 = \prod_{d|n} \Phi_d(x)$ for any positive integer n, and

$$x^{2i} - 1 = (x^i - 1)(1 + x^i),$$

it follows that

$$(1+x^i) = \prod_{\substack{d \mid 2i \\ d \nmid i}} \Phi_d(x). \tag{1}$$

Lemma 2.1.

(a) $1+x^i$ is irreducible in $\mathbb{Z}[x]$ if and only if $i = 2^k$ for some $k \geq 0$.

(b) $\gcd(1+x^n, 1+x^m) = \begin{cases} 1, & \text{if } \operatorname{val}_2(n) \neq \operatorname{val}_2(m) \\ 1+x^{\gcd(n,m)}, & \text{if } \operatorname{val}_2(n) = \operatorname{val}_2(m). \end{cases}$

(c) In particular, $(1+x^i) \mid (1+x^m)$ if and only if $m = i \cdot j$ for some odd j.

Proof. (a) The statement holds because the right hand side of (1) has a single factor if and only if $i = 2^k$ for some $k \geq 0$.

(b) From (1), it follows that

$$\gcd(1+x^n, 1+x^m) = \prod_{\substack{d \mid 2\cdot\gcd(n,m) \\ d \nmid n, d \nmid m}} \Phi_d(x). \tag{2}$$

Clearly, each d that occurs as an index in the product above satisfies $d \nmid \gcd(n,m)$.
Case 1: $\operatorname{val}_2(n) \neq \operatorname{val}_2(m)$. Assuming $n > m$, let $n = 2^j \cdot a \cdot \gcd(n,m)$ and $m = b \cdot \gcd(n,m)$, where $j > 0$ and a, b are odd. Then, if $d \mid 2 \cdot \gcd(n,m)$ we have $d \mid n$. Hence, the product in (2) is empty and $\gcd(1+x^n, 1+x^m) = 1$.
Case 2: $\operatorname{val}_2(n) = \operatorname{val}_2(m)$. Let $n = a \cdot \gcd(n,m)$ and $m = b \cdot \gcd(n,m)$, where a, b are odd. Then for each d such that $d \mid 2 \cdot \gcd(n,m)$ but $d \nmid \gcd(n,m)$ we have

$\mathrm{val}_2(d) = \mathrm{val}_2(\gcd(n,m)) + 1 = \mathrm{val}_2(n) + 1 = \mathrm{val}_2(m) + 1$, that is, $d \nmid n$ and $d \nmid m$. We may rewrite (2) as

$$\gcd(1+x^n, 1+x^m) = \prod_{\substack{d \mid 2\cdot\gcd(n,m) \\ d \nmid \gcd(n,m)}} \Phi_d(x) = 1 + x^{\gcd(n,m)}.$$

(c) Notice that $(1+x^i) \mid (1+x^m)$ is equivalent to $(1+x^i) = \gcd(1+x^i, 1+x^m)$, which by part (b) holds if and only if $\mathrm{val}_2(i) = \mathrm{val}_2(m)$ and $i = \gcd(i,m)$, that is, if and only if $m = i \cdot j$ for some odd j. □

Remark 2. If $i = \ell \cdot j$ with odd j, we have the factorization

$$1 + x^i = (1+x^\ell) \cdot \sum_{k=0}^{j-1} (-x^\ell)^k = (1+x^\ell)(x^{\ell(j-1)} - x^{\ell(j-2)} + \ldots - x^\ell + 1).$$

Corollary 2.2. *The* gcd *of any finite collection of products of terms of the form* $(1+x^i)$ *can be written as a product of terms of the form* $(1+x^j)$*. In particular,* $G(n,x)$ *decomposes into the product*

$$G(n,x) = \prod_{j=0}^{\lfloor \log_2(n) \rfloor} \gcd\{ \prod_{d \geq 1, d \text{ odd}} (1+x^{2^j d})^{\lfloor \frac{n}{2^j d} \rfloor - m_\lambda(2^j d)} \mid \lambda \vdash n \}.$$

We also recall basic facts about palindromic and unimodal polynomials.

Remark 3.

(a) Products and quotients of palindromic polynomials are palindromic and sums of palindromic polynomials of the same degree are palindromic. In particular, for all $i, \ell \geq 1$, the polynomial $(1+x^i)^\ell$ is palindromic.
(b) A product of palindromic, unimodal polynomials is again unimodal [2].

3. Ordinary partitions

Recall that

$$h_\lambda(x) = \prod_{i \geq 1} (1+x^i)^{\lfloor n/i \rfloor - m_\lambda(i)}.$$

By Lemma 2.1 (or also by Corollary 2.2) there exists a product decomposition of $G(n,x) = \gcd\{h_\lambda(x) \mid \lambda \vdash n\}$ as

$$G(n,x) = \prod_{i \geq 1} (1+x^i)^{c_{n,i}}\,. \tag{3}$$

Note that this is not a decomposition into coprime factors. Our next goal is to understand the exponents $c_{n,i}$. We first introduce more notation. Given a positive integer n, for each $i = 1, \ldots, n$, we define $\gamma_{n,i}$ as the maximal non-negative integer such that

$$(1+x^i)^{\gamma_{n,i}} \mid G(n,x)\,.$$

Clearly, $c_{n,i} \leq \gamma_{n,i}$.

The next lemma can be used to determine the coefficients $c_{n,i}$ recursively for fixed n.

Lemma 3.1. *With the notation above, for* $n \geq 1$ *we have*

(a) For all $i \geq 1$,

$$\gamma_{n,i} = \sum_{j \geq 1 \ odd} c_{n,i\cdot j} = \sum_{j>1 \ odd} \lfloor \frac{n}{i \cdot j} \rfloor .$$

(b) For $i > \lfloor \frac{n}{3} \rfloor$, $c_{n,i} = \gamma_{n,i} = 0$.

(c) For $3i > \lfloor \frac{n}{3} \rfloor \geq i$, $c_{n,i} = \gamma_{n,i}$.

Proof. These are consequences of Lemma 2.1: The term $(1 + x^i)$ divides $(1 + x^k)$ if and only if $k = i \cdot j$ for some odd $j \geq 1$, and if so, the multiplicity of $(1 + x^i)$ in the decomposition of $(1 + x^k)$ is one.

(a) For any i, we have $\gamma_{n,i} = \sum_{j \geq 1 \text{ odd}} c_{n,i\cdot j}$. On the other hand, since

$$h_\lambda(x) = \prod_{i \geq 1} (1 + x^i)^{\lfloor n/i \rfloor - m_\lambda(i)} ,$$

we also have

$$\gamma_{n,i} = \min_{\lambda \vdash n} \left(\sum_{j \text{ odd}} (\lfloor \frac{n}{i \cdot j} \rfloor - m_\lambda(i \cdot j)) \right) .$$

The sum on the right hand side is minimal for each λ such that $m_\lambda(i) = \lfloor \frac{n}{i} \rfloor$ and for such a λ we have $m_\lambda(i \cdot j) = 0$ for all $j > 1$. The minimality claim holds because, if λ has $m_\lambda(i) < \lfloor \frac{n}{i} \rfloor$, then

$$\sum_{j>1} m_\lambda(i \cdot j) < \sum_{j>1} j m_\lambda(i \cdot j) \leq \lfloor \frac{n}{i} \rfloor - m_\lambda(i).$$

Partitions λ with $m_\lambda(i) = \lfloor \frac{n}{i} \rfloor$ exist, for example $\lambda = (i^{\lfloor \frac{n}{i} \rfloor}, 1^{n - \lfloor \frac{n}{i} \rfloor})$. Hence, we obtain

$$\gamma_{n,i} = \sum_{j>1 \text{ odd}} \lfloor \frac{n}{i \cdot j} \rfloor.$$

(b) If $i > \lfloor n/3 \rfloor$, there is no multiple k of i satisfying $n \geq k = i \cdot j > i$ with j odd. So by (a), $0 = \sum_{j>1 \text{ odd}} \lfloor \frac{n}{i \cdot j} \rfloor = \gamma_{n,i} = c_{n,i}$.

(c) For $3i > \lfloor \frac{n}{3} \rfloor \geq i$ by (a) and (b), $\gamma_{n,i} = \sum_{j \geq 1 \text{ odd}} c_{n,i\cdot j} = c_{n,i}$. □

For each $i \geq 1$, $(1 + x^i)^{\gamma_{n,i}} \mid \text{num}^*(n, x)$ and $\gamma_{n,i}$ is the largest such power as shown below.

Lemma 3.2. *Let $p(x) = \prod_{i=1}^N (1 + x^i)^{a_i}$ and $q(x) = \prod_{i=1}^N (1 + x^i)^{b_i}$ be polynomials with exponents $a_i, b_i \geq 0$. Let α_i (resp. β_i) be the maximal exponent such that $(1 + x^i)^{\alpha_i} \mid p(x)$ (resp. $(1 + x^i)^{\beta_i} \mid q(x)$). If $\alpha_i = \beta_i$ for all i, then $p(x) = q(x)$.*

Proof. Because $\alpha_i = \sum_{j \geq 1 \text{ odd}} a_{i\cdot j}$ (resp. $\beta_i = \sum_{j \geq 1 \text{ odd}} b_{i\cdot j}$), and because $\alpha_i = a_i$ (resp. $\beta_i = b_i$) for $i > \lfloor N/3 \rfloor$, we obtain the coefficients a_i (resp. b_i) from the α_i (resp. β_i) by recursion. So if $\alpha_i = \beta_i$ for all i, then $a_i = b_i$ for all i. □

Using the results of the previous two lemmas, we prove the following about the quotients of $G(n, x)$ for consecutive values of n. We use the notation

$$\mathfrak{q}\mathrm{G}(n, x) := \frac{G(n, x)}{G(n - 1, x)} .$$

For a positive integer m, we denote by $\mathbf{o}(m)$ the largest odd divisor of m, that is,

$$\mathbf{o}(m) = \frac{m}{2^{\text{val}_2(m)}}.$$

Proposition 3.3. *Let $n > 1$. Then*

$$\mathfrak{q}\mathrm{G}(n,x) = \prod_{\substack{d|n \\ \mathbf{o}(n)\nmid d}} (1+x^d)\,,$$

In particular, if $n = 2^k$ then $\mathfrak{q}\mathrm{G}(n,x) = 1$.

Proof. Define

$$\chi(n,k) := \begin{cases} 1 & \text{if } k \mid n \\ 0 & \text{else,} \end{cases}$$

so that $\lfloor \frac{n}{k} \rfloor = \lfloor \frac{n-1}{k} \rfloor + \chi(n,k)$. By Lemma 3.1, for the maximal exponents $\gamma_{n,i}$ such that $(1+x^i)^{\gamma_{n,i}} \mid G(n,x)$ we have

$$\gamma_{n,i} = \sum_{j>1 \text{ odd}} \lfloor \frac{n}{i\cdot j} \rfloor = \gamma_{n-1,i} + \sum_{j>1 \text{ odd}} \chi(n, i\cdot j)\,.$$

For each $1 \le i \le n$, let $\delta_{n,i}$ be the maximal exponent such that $(1+x^i)^{\delta_{n,i}} \mid \prod_{\substack{d|n \\ \mathbf{o}(n)\nmid d}} (1+x^d)$. Then we have

$$\delta_{n,i} = \sum_{\substack{j\ge 1 \text{ odd} \\ i\cdot j \neq 2^{\mathrm{val}_2(i)-\mathrm{val}_2(n)} n}} \chi(n, i\cdot j) = \sum_{j>1 \text{ odd}} \chi(n, i\cdot j).$$

The last equality holds because both sides have the same number of summands with value 1 (and all other summands are zero).

So the maximal exponents $\eta_{n,i}$ such that $(1+x^i)^{\eta_{n,i}} \mid \prod_{\substack{d|n \\ \mathbf{o}(n)\nmid d}} (1+x^d)\cdot G(n-1,x)$ are

$$\eta_{n,i} = \delta_{n,i} + \gamma_{n-1,i} = \gamma_{n,i}\,.$$

By Lemma 3.2, we have $G(n,x) = \prod_{\substack{d|n \\ \mathbf{o}(n)\nmid d}} (1+x^d)\cdot G(n-1,x)$. □

Next, we examine the coefficients $c_{n,i}$ in

$$G(n,x) = \prod_{i=1}^{\lfloor \frac{n}{3} \rfloor} (1+x^i)^{c_{n,i}}.$$

Let $c_n := (c_{n,1}, c_{n,2}, \ldots, c_{n,n})$. By Lemma 3.1, we have $c_{n,i} = 0$ if $i > \lfloor \frac{n}{3} \rfloor$.

In the proof of our result we make use of the following identity for $b = 2$.

Lemma 3.4. *For all positive integers m and any integral base $b > 1$ we have*

$$m = \sum_{j=1,\, b\nmid j}^{m} \lfloor \log_b(\frac{bm}{j}) \rfloor.$$

Proof. We prove this statement by induction on m. When $m = 1$, both sides of the equation are 1. For the induction step notice that $\lfloor \log_b(\frac{b(m+1)}{j}) \rfloor = \lfloor \log_b(\frac{bm}{j}) \rfloor$ unless $\frac{b(m+1)}{j} = b^k$ for some $k > 0$. But there is a unique $j \le m+1$ with $b \nmid j$ that satisfies this condition. So, by the induction hypothesis,

$$\sum_{j=1,\, b\nmid j}^{m+1} \lfloor \log_b(\frac{b(m+1)}{j}) \rfloor = m+1\,.$$

□

Proposition 3.5. *Let n be a positive integer.*

(a) For $1 \le i \le n$,

$$c_{n,i} = \lfloor \frac{n}{i} \rfloor - \lfloor \log_2(\frac{n}{i}) \rfloor - 1 \,.$$

(b) $c_{n,i} > 0$ for $i = 1, \ldots, \lfloor n/3 \rfloor$, and c_n is weakly decreasing.

Proof. (a) We show that $\lfloor \frac{n}{i} \rfloor - \lfloor \log_2(\frac{n}{i}) \rfloor - 1$ satisfies (a)–(c) of Lemma 3.1. That is, $c_{n,i}$ and $\lfloor \frac{n}{i} \rfloor - \lfloor \log_2(\frac{n}{i}) \rfloor - 1$ satisfy the same recursion with the same initial conditions and thus must be equal. For $i > \lfloor \frac{n}{3} \rfloor$, obviously $\lfloor \frac{n}{i} \rfloor - \lfloor \log_2(\frac{n}{i}) \rfloor - 1 = 0$. Integers n, i satisfying $3i > \lfloor n/3 \rfloor \ge i$ are related by $n = ai + r$ with $0 \le r < i$ and $a \in \{3, 4, \ldots, 8\}$. For those

$$c_{n,i} = \gamma_{n,i} = \lfloor \frac{n}{3i} \rfloor + \lfloor \frac{n}{5i} \rfloor + \lfloor \frac{n}{7i} \rfloor = \lfloor \frac{a}{3} \rfloor + \lfloor \frac{a}{5} \rfloor + \lfloor \frac{a}{7} \rfloor \,,$$

and

$$\lfloor \frac{n}{i} \rfloor - \lfloor \log_2(\frac{n}{i}) \rfloor - 1 = a - \lfloor \log_2(a) \rfloor - 1 \,.$$

We check case-by-case the claim $c_{n,i} = a - \lfloor \log_2(a) \rfloor - 1$ holds for $a \in \{3, \ldots, 8\}$. So $\lfloor \frac{n}{i} \rfloor - \lfloor \log_2(\frac{n}{i}) \rfloor - 1$ satisfies (b) and (c) of Lemma 3.1. Next we check that it satisfies the identity in (a) of Lemma 3.1, that is, we show

$$\sum_{j \ge 1 \text{ odd}}^{\lfloor n/i \rfloor} (\lfloor \frac{n}{i \cdot j} \rfloor - \lfloor \log_2(\frac{n}{i \cdot j}) \rfloor - 1) = \sum_{j > 1 \text{ odd}} \lfloor \frac{n}{i \cdot j} \rfloor, \quad \text{for all} \quad n \ge i \,.$$

The above identity is equivalent to the following statement.

$$\lfloor \frac{n}{i} \rfloor = \sum_{j \ge 1 \text{ odd}}^{\lfloor \frac{n}{i} \rfloor} (\lfloor \log_2(\lfloor \frac{n}{i \cdot j} \rfloor) \rfloor + 1), \quad \text{for all} \quad n \ge i \,,$$

which is true by Lemma 3.4. This finishes the proof for (a).

(b) The statement follows immediately from (a). It can also be deduced from Lemma 3.1. □

Below, we show part of the infinite matrix $(c_{n,i})_{n,i \ge 1}$, where dots in rows represent zeros.

$$\begin{pmatrix}
. & . & . & . & . & . & . & . & \\
. & . & . & . & . & . & . & . & \\
1 & . & . & . & . & . & . & . & \\
1 & . & . & . & . & . & . & . & \\
2 & . & . & . & . & . & . & . & \\
3 & 1 & . & . & . & . & . & . & \\
4 & 1 & . & . & . & . & . & . & \ldots \\
4 & 1 & . & . & . & . & . & . & \\
5 & 1 & 1 & . & . & . & . & . & \\
6 & 2 & 1 & . & . & . & . & . & \\
7 & 2 & 1 & . & . & . & . & . & \\
8 & 3 & 1 & 1 & . & . & . & . & \\
9 & 3 & 1 & 1 & . & . & . & . & \\
10 & 4 & 1 & 1 & . & . & . & . & \\
 & & & \vdots & & & & &
\end{pmatrix}$$

Corollary 3.6. *The number of* 1*'s in* c_n *is* $\lfloor n/3 \rfloor - \lfloor n/5 \rfloor$.

More generally,

(a) if $j = 2^t - t - 1$ *for some* $t \in \mathbb{N}$*, then the number of entries equal to* j *in* c_n *is* $\lfloor \frac{n}{2^t-1} \rfloor - \lfloor \frac{n}{2^t+1} \rfloor$,

(b) if $j \neq 2^t - t - 1$ *for any* $t \in \mathbb{N}$*, then the number of entries equal to* j *in* c_n *is* $\lfloor \frac{n}{j+t_0} \rfloor - \lfloor \frac{n}{j+t_0+1} \rfloor$*, where* $t_0 \in \mathbb{N}$ *is such that* $2^{t_0} - t_0 - 1 < j < 2^{t_0+1} - (t_0 + 1) - 1$.

Proof. Using Proposition 3.5, one can check directly that $c_{n,i} = 1$ if and only if $\lfloor n/i \rfloor \in \{3, 4\}$, that is, $n/5 < i \leq n/3$. Then the number of 1's in c_n is $\lfloor n/3 \rfloor - \lfloor n/5 \rfloor$. The general statement is proved similarly. □

We denote by $\sigma_0(n)$ the number of divisors of n. Then, Proposition 3.5 leads to the following result on the number of factors of the form $(1 + x^i)$ in $G(n, x)$, counting multiplicities.

Corollary 3.7. *For positive integers* n*, we have*

$$\log_2(G(n, 1)) = \sum_{i=1}^{n} c_{n,i} = \left(\sum_{i=1}^{n} \sigma_0(i) \right) - \mathrm{val}_2(n!) - n.$$

Proof. This follows immediately from Proposition 3.5(a),

$$\sum_{i=1}^{n} \sigma_0(i) = \sum_{i=1}^{n} \lfloor \frac{n}{i} \rfloor \quad \text{and} \quad \sum_{i=1}^{n} \lfloor \log_2(\frac{n}{i}) \rfloor = \sum_{j=1}^{n} \mathrm{val}_2(j) = \mathrm{val}_2(n!)\,.$$

The identity above on the right is true because for all $k \geq 0$,

$$\#\{j \leq n \mid \mathrm{val}_2(j) = k\} = \#\{i \leq n \mid \lfloor \log_2(n/i) \rfloor = k\}\,.$$

□

Remark 4. Using Legendre's formula $\mathrm{val}_2(n!) = \sum_{i \geq 1} \lfloor n/2^i \rfloor$, we also have

$$\log_2(G(n, 1)) = \sum_{i \geq 1} (\lfloor \frac{n}{i} \rfloor - \lfloor \frac{n}{2^i} \rfloor - 1).$$

Proposition 3.8.

(a) $\mathrm{den}(n, x) = \prod_{i=1}^{n} (1 + x^i)^{a_{n,i}}$ *with* $a_{n,i} = \lfloor \log_2(\frac{n}{i}) \rfloor + 1$.

(b) The maximal exponents $\alpha_{n,i}$ *such that* $(1 + x^i)^{\alpha_{n,i}} \mid \mathrm{den}(n, x)$ *are* $\alpha_{n,i} = \lfloor \frac{n}{i} \rfloor$.

Proof. (a) By the definition of $\mathrm{den}(n, x)$ and Proposition 3.5,

$$a_{n,i} = \lfloor \frac{n}{i} \rfloor - c_{n,i} = \lfloor \log_2(\frac{n}{i}) \rfloor + 1\,.$$

(b) For $\mathrm{den}^*(n, x)$ the maximal exponents $\beta_{n,i}$ such that $(1 + x^i)^{\beta_{n,i}} \mid \mathrm{den}^*(n, x)$ are

$$\beta_{n,i} = \sum_{j \geq 1 \text{ odd}} \lfloor \frac{n}{i \cdot j} \rfloor = \gamma_{n,i} + \lfloor \frac{n}{i} \rfloor\,,$$

where $\gamma_{n,i}$ are the corresponding exponents of $G(n, x)$ (see Lemma 3.1). This determines the maximal exponents $\alpha_{n,i}$ such that $(1 + x^i)^{\alpha_{n,i}} \mid \mathrm{den}(n, x)$ are

$$\alpha_{n,i} = \beta_{n,i} - \gamma_{n,i} = \lfloor \frac{n}{i} \rfloor\,.$$

□

Proposition 3.9. *Let n be a positive integer. Then*

(a) $G(n,x)$, $\mathrm{num}(n,x)$, and $\mathrm{den}(n,x)$ are palindromic.
(b) $\mathrm{den}(n,x)$ and $G(n,x)$ are unimodal.

Proof. (a) Using Remark 3 (a), the statement follows from (3), Proposition 3.8 and the definition of $\mathrm{num}(n,x)$.

(b) For all m, the product $r(m,x)=\prod_{i=1}^{m}(1+x^i)$ is palindromic and unimodal (see e.g., [4], [6]). Recall that the exponents $a_{n,i}=\lfloor \log_2(\frac{n}{i})\rfloor+1$ of $\mathrm{den}(n,x)$ are weakly decreasing (Proposition 3.8 (a)). The same holds for the exponents $c_{n,i}$ of $G(n,x)$ (Proposition 3.5 (b)). So $\mathrm{den}(n,x)$, respectively $G(n,x)$, is the product of some $r(m,x)$. Then the statement follows from Remark 3. □

Next, we consider the quotient

$$\mathfrak{q}\mathrm{d}(n,x) := \frac{\mathrm{den}(n,x)}{\mathrm{den}(n-1,x)}.$$

Proposition 3.10. *Let $n\geq 2$. Then*

$$\mathfrak{q}\mathrm{d}(n,x) = \prod_{k=0}^{\mathrm{val}_2(n)} (1+x^{2^k\mathbf{o}(n)}) = \prod_{d|n}\Phi_{2d}(x)\,.$$

Moreover, $\deg \mathfrak{q}\mathrm{d}(n,x) = 2n-\mathbf{o}(n)$.

Proof. We have

$$\prod_{d|n}(1+x^d) = \frac{\mathrm{den}^*(n,x)}{\mathrm{den}^*(n-1,x)} = \mathfrak{q}\mathrm{G}(n,x)\cdot\mathfrak{q}\mathrm{d}(n,x).$$

Then,

$$\mathfrak{q}\mathrm{d}(n,x) = \prod_{k=0}^{\mathrm{val}_2(n)} (1+x^{2^k\mathbf{o}(n)}) = \prod_{d|n}\Phi_{2d}(x)\,,$$

where the first equality above follows from Proposition 3.3 and the second from (1).

The degree is

$$\deg \mathfrak{q}\mathrm{d}(n,x) = \sum_{k=0}^{\mathrm{val}_2(n)} 2^k\mathbf{o}(n) = 2n-\mathbf{o}(n).$$

□

There are various ways to calculate the degree of $\mathrm{den}(n,x)$, which leads to several equivalent formulas that, in turn, yield nice sum identities:

Let $s(n)$ be the sequence defined recursively for $n\geq 1$ by

$$s(2n)=s(n)+2n \quad \text{and} \quad s(2n-1)=2n-1.$$

This is sequence A129527 in [5]. This sequence also appears in the generating function for the number $\mathrm{PL}(n)$ of plane partitions of n (see [3]):

$$\sum_{n\geq 0}\mathrm{PL}(n)x^n = \prod(1+x^n)^{s(n)}.$$

Corollary 3.11. *For $n\geq 2$,* $\deg \mathfrak{q}\mathrm{d}(n,x) = s(n)$.

Proof. By Proposition 3.10, $\deg \mathfrak{qd}(2m+1,x) = 2m+1$ and

$$\deg \mathfrak{qd}(2m,x) = 4m - \mathbf{o}(m) = \deg \mathfrak{qd}(m,x) + 2m\,.$$

□

Proposition 3.12. *Let $n \geq 1$. Then*

$$\deg \operatorname{den}(n,x) = \sum_{i=1}^{n} s(i) = n(n+1) - \sum_{i=1}^{n} \mathbf{o}(i) = \frac{n(n+1)}{2} + \sum_{i=1}^{n} i\lfloor \log_2(\frac{n}{i})\rfloor\,;$$

$$\deg \operatorname{num}(n,x) = n^2 - \sum_{i=1}^{n} \mathbf{o}(i) = \frac{n(n-1)}{2} + \sum_{i=1}^{n} i\lfloor \log_2(\frac{n}{i})\rfloor\,.$$

Proof. Since

$$\deg \operatorname{den}(n,x) = 1 + \sum_{i=2}^{n} \deg \mathfrak{qd}(i,x)\,,$$

the first two equalities for $\deg \operatorname{den}(n,x)$ follow from Corollary 3.11 and Proposition 3.10 respectively. For the third equality, we use Proposition 3.8 (a). Finally, by definition $\deg \operatorname{num}(n,x) = \deg \operatorname{den}(n,x) - n$. □

Remark 5. We can now see that Conjecture 1 implies Conjecture 2. If $\operatorname{num}(n,x)$ is irreducible, then $\gcd\big(\operatorname{num}(n,x), \operatorname{den}(n,x)\big)$ must be either 1 or $\operatorname{num}(n,x)$. However, $\operatorname{den}(n,x)$ is a ratio of $(1+x^i)$'s, thus its irreducible factors are among the cyclotomic polynomials Φ_{2d}, $d \leq n$. Hence, if $\gcd\big(\operatorname{num}(n,x), \operatorname{den}(n,x)\big) = \operatorname{num}(n,x)$, then $\operatorname{num}(n,x) = \Phi_{2d}$ for some $d \leq n$, which contradicts the degree considerations of Proposition 3.12.

Proposition 3.13. *For all $n \geq 1$,*

(a) $\operatorname{den}(n,1) = 2^{\operatorname{val}_2((2n)!)}$,

(b) $\operatorname{num}(n,1) = 2^{\operatorname{val}_2(n!)} P(n)$.

Proof. (a) For $n = 1$ the claim is obvious. For $n > 1$ we have the recursion $\operatorname{den}(n,x) = \mathfrak{qd}(n,x)\operatorname{den}(n-1,x)$. Then, using induction and the explicit formula in Proposition (3.10), we have

$$\operatorname{den}(n,1) = \mathfrak{qd}(n,1)\operatorname{den}(n-1,1) = 2^{\operatorname{val}_2(n)+1} \cdot 2^{\operatorname{val}_2((2n-2)!)}.$$

Since $\operatorname{val}_2(n) + 1 = \operatorname{val}_2(2n) + \operatorname{val}_2(2n-1)$, the claim follows.

(b) For any λ, we have $h_\lambda(1) = 2^{\sum_{i=1}^{n} \lfloor n/i \rfloor - n}$. Then, using Corollary 3.7, we have

$$\operatorname{num}(n,1) = \frac{P(n) \cdot 2^{\sum_{i=1}^{n} \lfloor n/i \rfloor - n}}{G(n,1)} = P(n) \cdot 2^{\operatorname{val}_2(n!)}\,.$$

□

For the remainder of this section we study $\operatorname{num}(n,x)$, in particular, its irreducibility and values at roots of unity. We set $\operatorname{num}(0,x) := 1$ and give $\operatorname{num}(n,x)$ explicitly for $1 \leq n \leq 5$.

Example 1. Let $D(n) = \deg \operatorname{num}(n,x)$. For small n we list $\operatorname{num}(n,x) = \sum_{k=0}^{D(n)} a_k x^k$ by giving its sequence of coefficients.

n	$D(n)$	$(a_0, \ldots, a_{D(n)})$
1	0	(1)
2	2	(2, 2, 2)
3	4	(3, 2, 4, 2, 3)
4	10	(5, 8, 15, 14, 24, 20, 24, 14, 15, 8, 5)
5	15	(7, 9, 21, 14, 37, 21, 51, 24, 51, 21, 37, 14, 21, 9, 7)

It can be checked directly that for $1 \le n \le 5$, the polynomial $\mathrm{num}(n, x)$ is irreducible over the integers, and in particular $\mathrm{num}(n, \zeta_{2d}) \neq 0$ for any d. Here, ζ_j is a primitive jth root of unity.

We write (a^m, μ) for a partition with largest part equal to a and repeated m times, and the remaining smaller parts forming a partition μ. Given a partition λ, we define

$$q_\lambda(x) := \frac{h_\lambda(x)}{G(n,x)}\,.$$

Then $\mathrm{num}(n, x) = \sum_{\lambda \vdash n} q_\lambda(x)$.

To simplify the arguments, we introduce the following notation. For positive integers n, d, and remainder $r = n - d\lfloor \frac{n}{d} \rfloor$, we define

$$\begin{aligned} f(n,d,x) &:= \frac{\prod_{i \neq d}(1+x^i)^{\lfloor n/i \rfloor}}{G(n,x)} \cdot \frac{G(r,x)}{\prod_i (1+x^i)^{\lfloor r/i \rfloor}} \\ &= \frac{\mathrm{den}(n,x)}{(1+x^d)^{\lfloor n/d \rfloor}\mathrm{den}(r,x)}\,. \end{aligned} \tag{4}$$

To further examine $\mathrm{num}(\zeta_{2d})$, we prove two useful lemmas.

Lemma 3.14. *Fix $n \ge 1, d \ge 1$ and $\lambda \vdash n$. Let $r = n - d\lfloor \frac{n}{d} \rfloor$. Then*

(a) $(1+x^d) \nmid q_\lambda(x)$ if and only if $\lambda = (d^{\lfloor n/d \rfloor}, \mu)$, where $\mu \vdash r$. Thus

$$\mathrm{num}(n, \zeta_{2d}) = \sum_{\substack{\lambda = (d^{\lfloor n/d \rfloor}, \mu) \\ \text{with } \mu \vdash r}} q_\lambda(\zeta_{2d})\,.$$

(b) $(1+x^d) \nmid q_\lambda(x)$ if and only if $\Phi_{2d}(x) \nmid q_\lambda(x)$.

(c) $\mathrm{num}(n, \zeta_{2d}) = f(n, d, \zeta_{2d})\,\mathrm{num}(r, \zeta_{2d})$.

(d) $\mathrm{num}(n, \zeta_{2d}) \neq 0$ if and only if $\mathrm{num}(r, \zeta_{2d}) \neq 0$.

Proof. (a) By Lemma 3.1, $(1+x^d)^k \mid q_\lambda(x)$ if and only if

$$\begin{aligned} k &\le \sum_{j \ge 1,\ \text{odd}} (\lfloor \frac{n}{d \cdot j} \rfloor - m_\lambda(d \cdot j)) - \sum_{j > 1,\ \text{odd}} \lfloor \frac{n}{d \cdot j} \rfloor \\ &= \lfloor \frac{n}{d} \rfloor - \sum_{j \ge 1,\ \text{odd}} m_\lambda(d \cdot j). \end{aligned}$$

Since $\lfloor \frac{n}{d} \rfloor = \sum_{j \ge 1,\ \text{odd}} m_\lambda(d \cdot j)$ if and only if $\lfloor \frac{n}{d} \rfloor = m_\lambda(d)$, the result follows.

(b) It follows from Remark 1 and Lemma 2.1 that

$$\Phi_{2d} \mid (1+x^i) \quad \Longleftrightarrow \quad (1+x^d) \mid (1+x^i)\,.$$

Since $q_\lambda(x) = \frac{h_\lambda(x)}{G(n,x)}$ is a ratio of products of polynomials of the form $(1+x^i)$, the result follows.

(c) First note that if $\lambda = (d^{\lfloor n/d \rfloor}, \mu)$, then

$$q_\lambda(x) = f(n,d,x) \cdot q_\mu(x).$$

Hence,

$$\sum_{\substack{\lambda=(d^{\lfloor \frac{n}{d} \rfloor},\mu) \\ \text{with } \mu \vdash r}} q_\lambda(x) = f(n,d,x) \cdot \operatorname{num}(r,x)$$

and the result follows from (a).

(d) By (a), $f(n,d,\zeta_{2d}) \neq 0$ and the result follows from (c). □

We deduce a partial result related to Conjecture 2:

Corollary 3.15. *Fix $n \geq 1$ and $d \geq 1$, and let $r = n - d\lfloor \frac{n}{d} \rfloor$. Then*

$$\Phi_{2d}(x) \nmid \operatorname{num}(n,x)$$

in each of the following cases:

(a) $r \in \{0,1,\dots,5\}$,

(b) $\deg \operatorname{num}(r,x) < \varphi(2d)$, where φ is Euler's totient function.

In particular, in each of the above cases $\Phi_{2d}(x) \nmid \gcd(\operatorname{num}(n,x), \operatorname{den}(n,x))$.

Proof. We apply Lemma 3.14 (d).

(a) Example 1 shows that $\operatorname{num}(r,\zeta_{2d}) \neq 0$ for each such r.

(b) If $\deg \operatorname{num}(r,x) < \varphi(2d) = \deg \Phi_{2d}(x)$, then $\operatorname{num}(r,\zeta_{2d}) \neq 0$. □

Concerning Conjecture 2, $\gcd(\operatorname{num}(n,x), \operatorname{den}(n,x))$ is at most a product of $\Phi_{2d} \mid (1+x^i)$, $i \leq n$. Moreover, Corollary 3.15 excludes some Φ_{2d}. It is natural to search for zeros of $\operatorname{num}(n,x)$ on the unit circle, that is, to use methods similar to the circle method. Some experimental evidence shows that there are many roots of $\operatorname{num}(n,x)$ on the unit circle. However, we could not find any ζ_{2d} among them. This can already be observed for small n from Example 1: the roots of $\operatorname{num}(n,x)$ are on the unit circle for $n \leq 3$, the roots of $\operatorname{num}(4,x)$ are not on the unit circle, and ten zeros of $\operatorname{num}(5,x)$ are on the unit circle.

In the proof of the next lemma, we make use of the identity

$$\prod_{k=0}^{\operatorname{val}_2(m)} (1 + x^{2^k \mathbf{o}(m)}) = \frac{1-x^{2m}}{1-x^{\mathbf{o}(m)}}\,.$$

Lemma 3.16. *Let $n \geq d \geq 1$. Then*

$$\frac{f(n,d,\zeta_{2d})}{f(n-d,d,\zeta_{2d})} = d\lfloor \frac{n}{d} \rfloor \cdot \frac{2}{\prod_{j=0}^{d-1}(1-\zeta_{2d}^{\mathbf{o}(n-j)})}\,.$$

Proof. Using (4) and Proposition 3.10, we obtain

$$\begin{aligned} \frac{f(n,d,x)}{f(n-d,d,x)} &= \frac{1}{(1+x^d)} \frac{\operatorname{den}(n,x)}{\operatorname{den}(n-d,x)} \\ &= \frac{1}{(1+x^d)} \prod_{j=0}^{d-1} \prod_{k=0}^{\operatorname{val}_2(n-j)} (1 + x^{2^k \mathbf{o}(n-j)})\,. \end{aligned}$$

Among all the factors $(1+x^{2^k \mathbf{o}(n-j)})$ in the double product, there is a single factor with $2^k \mathbf{o}(n-j)$ equal to an odd multiple of d, namely $2^{\operatorname{val}_2(d)} \mathbf{o}(n-r) = d \cdot \mathbf{o}(\lfloor n/d \rfloor)$. Moreover, this factor occurs only once. We have

$$\frac{1+x^{d \cdot \mathbf{o}(\lfloor n/d \rfloor)}}{1+x^d} = 1 - x^d + x^{2d} - \cdots + x^{(\mathbf{o}(\lfloor n/d \rfloor)-1)d}.$$

Then,

$$1-\zeta_{2d}^{d}+\zeta_{2d}^{2d}-\cdots+\zeta_{2d}^{(\mathbf{o}(\lfloor n/d\rfloor)-1)d}=\mathbf{o}(\lfloor n/d\rfloor).$$

We evaluate at ζ_{2d} the factor of $\frac{f(n,d,x)}{f(n-d,d,x)}$ corresponding to $j=r=n-d\lfloor n/d\rfloor$ and obtain

$$\begin{aligned}
&\frac{1}{(1+x^d)}\prod_{k=0}^{\mathrm{val}_2(d\lfloor n/d\rfloor)}(1+x^{2^k\mathbf{o}(d\lfloor n/d\rfloor)})\Bigg|_{x=\zeta_{2d}}\\
&\quad=\mathbf{o}(\lfloor n/d\rfloor)\prod_{\substack{k=0\\k\neq\mathrm{val}_2(d)}}^{\mathrm{val}_2(d\lfloor n/d\rfloor)}(1+\zeta_{2d}^{2^k\mathbf{o}(d\lfloor n/d\rfloor)})\\
&\quad=\lfloor n/d\rfloor\prod_{k=0}^{\mathrm{val}_2(d)-1}(1+\zeta_{2d}^{2^k\mathbf{o}(d\lfloor n/d\rfloor)})\\
&\quad=\lfloor n/d\rfloor\frac{1-\zeta_{2d}^{2^{\mathrm{val}_2(d)}\mathbf{o}(d\lfloor n/d\rfloor)}}{1-\zeta_{2d}^{\mathbf{o}(d\lfloor n/d\rfloor)}}=\lfloor n/d\rfloor\frac{2}{1-\zeta_{2d}^{\mathbf{o}(d\lfloor n/d\rfloor)}}\,.
\end{aligned}$$

The product of the remaining factors is

$$\prod_{\substack{j=0\\j\neq r}}^{d-1}\prod_{k=0}^{\mathrm{val}_2(n-j)}(1+\zeta_{2d}^{2^k\mathbf{o}(n-j)})=\prod_{\substack{j=0\\j\neq r}}^{d-1}\frac{1-\zeta_{2d}^{2(n-j)}}{1-\zeta_{2d}^{\mathbf{o}(n-j)}}\,.$$

Notice that

$$\prod_{j=0,j\neq r}^{d-1}(1-\zeta_{2d}^{2(n-j)})=\prod_{\substack{j=0\\j\neq r}}^{d-1}(1-\zeta_d^{(n-j)})=\prod_{j=0}^{d-1}(1-\zeta_d^j)=\left.\frac{x^d-1}{x-1}\right|_{x=1}=d.$$

This completes the proof. □

Proposition 3.17. *For $n,d\geq 1$ and $r=n-d\lfloor n/d\rfloor$,*

$$\mathrm{num}(n,\zeta_{2d})=\frac{(2d)^{\lfloor n/d\rfloor}\lfloor\frac{n}{d}\rfloor!}{\prod_{j=r+1}^{n}(1-\zeta_{2d}^{\mathbf{o}(j)})}\cdot\mathrm{num}(r,\zeta_{2d})\,.$$

Proof. Since $f(r,d,x)=1$, it follows that

$$f(n,d,x)=\prod_{j=1}^{\lfloor n/d\rfloor}\frac{f(r+jd,d,x)}{f(r+(j-1)d,d,x)}.$$

By Lemma 3.14(c), we have $\mathrm{num}(n,\zeta_{2d})=f(n,d,\zeta_{2d})\,\mathrm{num}(r,\zeta_{2d})$. The claim then follows from Lemma 3.16. □

Theorem 3.18. *For all $n\geq 1$,*

(a) $\mathrm{num}(n,-1)=n!$

(b) $|\mathrm{num}(n,\zeta_4)|=|\mathrm{num}(n,\pm\mathrm{i})|=2^{\lfloor n/2\rfloor}\lfloor n/2\rfloor!$

(c) $|\mathrm{num}(n,\zeta_6)|=\begin{cases}3^{\lfloor n/3\rfloor}\lfloor n/3\rfloor! & n\not\equiv 2\pmod 3\\ 4\cdot 3^{\lfloor n/3\rfloor}\lfloor n/3\rfloor! & n\equiv 2\pmod 3.\end{cases}$

Proof. We apply Proposition 3.17. Recall $\mathrm{num}(0,x) = \mathrm{num}(1,x) = 1$ and $\mathrm{num}(2,x) = 2(1+x+x^2)$.

(a) For $d=1$ we find $\frac{f(j,1,-1)}{f(j-1,1,-1)} = j$, so $\mathrm{num}(n,-1) = n!$.

(b) For $d=2$ we have

$$\frac{f(r+2j,2,\zeta_4)}{f(r+2(j-1),2,\zeta_4)} = 2j\frac{2}{(1-\zeta_4^{\mathbf{o}(r+2j)})(1-\zeta_4^{\mathbf{o}(r+2j-1)})}.$$

Since $|1-\zeta_4^{\mathrm{odd}}| = \sqrt{2}$, we obtain $|\mathrm{num}(n,\zeta_4)| = 2^{\lfloor n/2\rfloor}\lfloor n/2\rfloor!$ as claimed.

(c) For $d=3$ we have

$$\frac{f(r+3j,3,\zeta_6)}{f(r+3(j-1),3,\zeta_6)} = 3j\frac{2}{(1-\zeta_6^{\mathbf{o}(r+3j)})(1-\zeta_6^{\mathbf{o}(r+3j-1)})(1-\zeta_6^{\mathbf{o}(r+3j-2)})}.$$

Next, note that $\mathbf{o}(n) \equiv 3 \pmod 6$ if and only if $n \equiv 0 \pmod 3$. This implies $|(1-\zeta_6^{\mathbf{o}(r+3j)})(1-\zeta_6^{\mathbf{o}(r+3j-1)})(1-\zeta_6^{\mathbf{o}(r+3j-2)})| = 2$. Thus, $|\frac{f(r+3j,3,\zeta_6)}{f(r+3(j-1),3,\zeta_6)}| = 3j$, and the claim follows by noticing that $|\mathrm{num}(2,\zeta_6)| = 4$. □

In general, the expression in Proposition 3.17 does not simplify as nicely as in Theorem 3.18.

We conclude this section with the following two conjectures for which we have experimental evidence.

Conjecture 3. *Write* $\mathrm{num}(n,x) = \mathrm{num}_0(n,x) + \mathrm{num}_1(n,x)$ *as the sum of an even and an odd function. Then* $\mathrm{num}_0(n,x)$ *is unimodal.*

Conjecture 4. *The sequence of coefficients of* $\mathrm{den}(n,x)$ *is log-concave except for* $n = 3,5,6,7$.

4. Binary partitions

Let $\mathcal{B}(n)$ denote the set of binary partitions of n, that is, partitions whose parts are powers of 2. We set $B(n) = |\mathcal{B}(n)|$.

We define

$$\mathrm{sr}_{\mathcal{B}}(n,x) := \sum_{\lambda\in\mathcal{B}(n)} \frac{1}{\mathrm{sp}(\lambda,x)}.$$

We have

$$\mathrm{sr}_{\mathcal{B}}(n,x) = \frac{\mathrm{num}^*_{\mathcal{B}}(n,x)}{\mathrm{den}^*_{\mathcal{B}}(n,x)},$$

where

$$\mathrm{num}^*_{\mathcal{B}}(n,x) = \prod_{i=0}^{\lfloor \log_2(n)\rfloor} (1+x^{2^i})^{\lfloor n/2^i\rfloor - m_\lambda(2^i)}$$

and

$$\mathrm{den}^*_{\mathcal{B}}(n,x) = \prod_{i=0}^{\lfloor \log_2(n)\rfloor} (1+x^{2^i})^{\lfloor n/2^i\rfloor}.$$

If λ is a binary partition of n, we let

$$h_{\mathcal{B},\lambda}(x) := \prod_{i=0}^{\lfloor \log_2(n)\rfloor} (1+x^{2^i})^{\lfloor n/2^i\rfloor - m_\lambda(2^i)}.$$

Throughout this section, for ease of notation, we let $h_\lambda(x) = h_{\mathcal{B},\lambda}(x)$.

Then

$$\mathrm{num}^*_{\mathcal{B}}(n,x) = \sum_{\lambda\in\mathcal{B}(n)} h_\lambda(x) =: \sum_{k=0}^{d(n)} b_k(n)x^k.$$

We set $\mathrm{num}^*_{\mathcal{B}}(0,x) := 1$. For all $\lambda \in \mathcal{B}(n)$,

$$d(n) := \deg \mathrm{num}^*_{\mathcal{B}}(n,x) = \deg h_\lambda(x).$$

Remark 6. We note that

$$G_{\mathcal{B}}(n,x) := \gcd\{h_\lambda(x) \mid \lambda \in \mathcal{B}(n)\} = 1.$$

For $n = 1$ the statement is trivial. If $n \geq 2$, $(1+x) \mid h_{(2,1^{n-1})}(x)$ but $(1+x) \nmid h_{(1^n)}(x)$. If $1 \leq i \leq \lfloor \log_2(n) \rfloor$, then $(1+x^{2^i}) \mid h_{(1^n)}(x)$ but $(1+x^{2^i}) \nmid h_\mu(x)$, where μ is a partition with $m_\mu(2^i) = \lfloor \frac{n}{2^i} \rfloor$. The result follows from Lemma 2.1.

In analogy with the ordinary partitions, for the remainder of the section we use

$$\mathrm{num}_{\mathcal{B}}(n,x) := \frac{\mathrm{num}^*_{\mathcal{B}}(n,x)}{G_{\mathcal{B}}(n,x)} = \mathrm{num}^*_{\mathcal{B}}(n,x)\,,$$

$$\mathrm{den}_{\mathcal{B}}(n,x) := \frac{\mathrm{den}^*_{\mathcal{B}}(n,x)}{G_{\mathcal{B}}(n,x)} = \mathrm{den}^*_{\mathcal{B}}(n,x)\,.$$

We have the following main conjecture for binary partitions.

Conjecture 5. *For $n \geq 1$,*

$$\gcd(\mathrm{num}_{\mathcal{B}}(n,x), \mathrm{den}_{\mathcal{B}}(n,x)) = 1.$$

We begin by proving several useful properties of $\mathrm{num}_{\mathcal{B}}(n,x)$. Here and throughout, if $\mu \subseteq \lambda$ as multisets, we write $\lambda \setminus \mu$ for the partition obtained by removing the parts of μ from λ. Moreover, if $\lambda \in \mathcal{B}(n)$ has $m_\lambda(1) = 0$, we denote by $\lambda/2$ the partition contained from λ by halving each part.

Proposition 4.1. *Let $n \in \mathbb{N}$. Then*

(a) $\mathrm{num}_{\mathcal{B}}(2n+1,x) = \mathrm{num}_{\mathcal{B}}(2n,x)$.

(b) For all n,

$$d(n) = \sum_{j\geq 1} 2^j \cdot \lfloor \frac{n}{2^j} \rfloor.$$

(c) $d(n)$ is even and $2^{\lfloor \log_2(n)\rfloor}\lfloor \log_2(n)\rfloor \leq d(n) \leq n\lfloor \log_2(n)\rfloor$. More precisely,

$$d(n) = n\lfloor \log_2(n)\rfloor - \sum_{j=\mathrm{val}_2(n)+1}^{\lfloor \log_2(n)\rfloor} r_j,$$

where r_j is the remainder of n on division by 2^j.

(d) $b_k(n) > 0$ for all $k = 0, \ldots, d(n)$.

(e) $\mathrm{num}_{\mathcal{B}}(n,x)$ *and* $\mathrm{den}_{\mathcal{B}}(n,x)$ *are palindromic.*

(f) The first coefficients of $\mathrm{num}_{\mathcal{B}}(n,x)$ *are*

$$b_0(n) = B(n), \qquad b_1(n) = \sum_{\lambda\in\mathcal{B}(n)} (n - m_\lambda(1)),$$

$$b_2(n) = \sum_{\lambda\in\mathcal{B}(n)} \left(\binom{n-m_\lambda(1)}{2} + \lfloor n/2 \rfloor - m_\lambda(2) \right).$$

(g) $\mathrm{num}_{\mathcal{B}}(n,-1) = 2^{\mathrm{val}_2(n!)}$ *for $n > 0$.*

(h) *For $n > 1$, $(1 + x + x^2) \mid \mathrm{num}_{\mathcal{B}}(n, x)$, that is, the primitive third roots of unity are roots of $\mathrm{num}_{\mathcal{B}}(n, x)$.*

Proof. (a) If $\lambda \in \mathcal{B}(2n+1)$, then $m_\lambda(1) \geq 1$. The transformation $\lambda \to \lambda \setminus (1)$ is a bijection from $\mathcal{B}(2n+1)$ to $\mathcal{B}(2n)$. We verify $h_{\lambda\setminus(1)}(x) = h_\lambda(x)$, which completes the proof.

(b) If $\lambda \in \mathcal{B}(n)$, $d(n) = \deg h_\lambda(x)$ and

$$\deg h_\lambda(x) = \sum_{i \geq 0} 2^i \cdot (\lfloor n/2^i \rfloor - m_\lambda(2^i)) = \sum_{i \geq 1} 2^i \cdot \lfloor n/2^i \rfloor.$$

(c) Follows immediately from (b).

(d) If k is even, x^k occurs in $h_{(1^n)}(x)$. If k is odd, x^k occurs in $h_{(2,1^{n-2})}(x)$.

(e) Follows from Remark 3.

(f) Let $\lambda \in \mathcal{B}(n)$ and denote the coefficient of x^i in $h_\lambda(x)$ by $[x^i]h_\lambda$. Then,

$$\begin{aligned}
[x^0]h_\lambda &= 1 \\
[x^1]h_\lambda &= \binom{\lfloor n/2^0 \rfloor - m_\lambda(2^0)}{1} = n - m_\lambda(1) \\
[x^2]h_\lambda &= \binom{\lfloor n/2^0 \rfloor - m_\lambda(2^0)}{2} + \binom{\lfloor n/2^1 \rfloor - m_\lambda(2^1)}{1}.
\end{aligned}$$

Summing over all partitions in $\mathcal{B}(n)$ completes the proof.

(g) Clearly, $\mathrm{num}_{\mathcal{B}}(1, -1) = 1$. If $n > 1$, $\lambda \in \mathcal{B}(n)$ and $\lambda \neq (1^n)$, then $h_\lambda(-1) = 0$. Hence, for $n > 1$,

$$\mathrm{num}_{\mathcal{B}}(n, -1) = h_{(1^n)}(-1) = \prod_{i \geq 1} 2^{\lfloor n/2^i \rfloor} = 2^{\sum_{i \geq 1} \lfloor n/2^i \rfloor}.$$

Since $\lfloor n/2^i \rfloor$ is the number of positive integers $m \leq n$, such that $2^i \mid m$, it follows that $\sum_{i \geq 1} \lfloor n/2^i \rfloor = \mathrm{val}_2(n!)$.

(h) Let $n > 1$. We first make the following observation. If $i > 1$ and $\lambda \in \mathcal{B}(n)$ satisfies $m_\lambda(2^i) > 0$, let $\lambda^{(i)}$ be the partition obtained from λ by removing a part equal to 2^i and inserting two parts equal to 2^{i-1}. Then

$$\begin{aligned}
h_\lambda(x) + h_{\lambda^{(i)}}(x) = \prod_{j \neq i, i-1} & (1 + x^{2^j})^{\lfloor n/2^j \rfloor - m_\lambda(2^j)} \\
& \times (1 + x^{2^i})^{\lfloor n/2^i \rfloor - m_\lambda(2^i)} \cdot (1 + x^{2^{i-1}})^{\lfloor n/2^{i-1} \rfloor - m_\lambda(2^{i-1}) - 2} \\
& \times \left((1 + x^{2^{i-1}})^2 + (1 + x^{2^i}) \right).
\end{aligned}$$

The last factor $\left((1 + x^{2^i}) + (1 + x^{2^{i-1}})^2 \right) = 2(1 + x^{2^{i-1}} + x^{2^i})$ is divisible by $(1 + x + x^2)$. To see this, note that $\{2^i \pmod 3, 2^{i-1} \pmod 3\} \equiv \{1, 2\}$ and thus the primitive third roots of unity ζ_3, ζ_3^2 are roots of $(1 + x^{2^{i-1}} + x^{2^i})$. Hence, $(1 + x + x^2) \mid (h_\lambda(x) + h_{\lambda^{(i)}}(x))$. To complete the proof we write $\mathcal{B}(n)$ as a disjoint union of pairs $\lambda, \lambda^{(i)}$. Note that i can vary among pairs. We proceed as follows.

First, we decompose $\mathcal{B}(n)$ into the disjoint subsets $\mathcal{B}^{(i)}(n)$, $i = 0, \ldots \lfloor \log_2(n) \rfloor$,

$$\mathcal{B}^{(i)}(n) = \{\lambda \in \mathcal{B}(n) \mid \lambda_1 = 2^i\}.$$

Thus, $\lambda \in \mathcal{B}^{(i)}(n)$ if and only if $m_\lambda(2^i) \geq 1$ and $m_\lambda(2^j) = 0$ for $j > i$. We decompose each $\mathcal{B}^{(i)}(n)$ into the disjoint union of subsets $\mathcal{B}^{(i,m)}(n)$, $m = 1, \ldots \lfloor \frac{n}{2^i} \rfloor$.

$$\mathcal{B}^{(i,m)}(n) = \{\lambda \in \mathcal{B}^{(i)}(n) \mid m_\lambda(2^i) = m\}.$$

We create pairs $\lambda, \lambda^{(i)}$ using the following algorithm.

(0) Set $\mathcal{B} = \mathcal{B}(n)$, $i = \lfloor \log_2(n) \rfloor$, $m = \lfloor \frac{n}{2^i} \rfloor$.
(1) As long as $\mathcal{B}^{(i,m)} \cap \mathcal{B} \neq \emptyset$, choose a $\lambda \in \mathcal{B}^{(i,m)} \cap \mathcal{B}$ and put $\mathcal{B} = \mathcal{B} \setminus \{\lambda, \lambda^{(i)}\}$.
(2) If $\mathcal{B} = \emptyset$ STOP. Else put $m = m - 1$.
(3) If $m \geq 1$ go to (1). Else put $i = i - 1$, $m = \lfloor \frac{n}{2^i} \rfloor$, and go to (1).

The algorithm terminates when $i = 1$. □

Related to Proposition 4.1(e) we conjecture the following behavior of the coefficients of $\mathrm{num}_{\mathcal{B}}(n, x)$.

Conjecture 6. *For $n \geq 2$, $\mathrm{num}_{\mathcal{B}}(n, x)$ is unimodal and log-concave.*

The reformulation of $b_1(n)$ below follows from Proposition 4.1 (f).

Corollary 4.2. *If $n \geq 1$, then $b_1(n) = nB(n) - \sum_{k=0}^{n-1} B(k)$.*

Proof. From Proposition 4.1(f), we have

$$b_1(n) = nB(n) - \sum_{\lambda \in \mathcal{B}(n)} m_\lambda(1).$$

Recall the standard combinatorial argument used in the proof of Stanley's Theorem: if $\lambda \in \mathcal{B}(n)$ with $m_\lambda(1) \geq j$, the jth occurrence of 1 in λ corresponds to the partition $\lambda \setminus (1^j) \in \mathcal{B}(n - j)$. This shows that

$$\sum_{\lambda \in \mathcal{B}(n)} m_\lambda(1) = \sum_{k=0}^{n-1} B(k). \tag{5}$$

□

Remark 7. The sequence $nB(n)$ is sequence $A304909$ in [5] and gives the sum of all parts in all partitions in $\mathcal{B}(n)$. Thus, $b_1(n)$ is also the sum of all parts greater than 1 in all partitions in $\mathcal{B}(n)$.

We continue with results about the linear and quadratic coefficients, $b_1(n)$ and $b_2(n)$, of $\mathrm{num}_B(n, x)$. For a sequence s_n we denote by Δs_n the sequence of forward differences of s_n, that is,

$$\Delta s_n = s_{n+1} - s_n.$$

Proposition 4.3. *Let $n \geq 0$. Then*

$$\Delta b_1(n) = \begin{cases} 0 & \text{if } n \text{ even} \\ (n+1)B(\frac{n+1}{2}) & \text{if } n \text{ odd.} \end{cases}$$

Proof. If $n = 0$ the statement is clearly true. Let $n \geq 1$. By Proposition 4.1(a), if $n = 2j$ for some $j \geq 1$,

$$b_1(2j+1) - b_1(2j) = 0.$$

If $n = 2j + 1$ for some $j \geq 0$, by Corollary 4.2,

$$b_1(2j+2) - b_1(2j+1) = (2j+2)\left(B(2j+2) - B(2j+1)\right).$$

The transformation $\lambda \to \lambda \setminus (1)$ is a bijection from $\{\lambda \in \mathcal{B}(2j+2) \mid m_\lambda(1) \geq 1\}$ to $\mathcal{B}(2j+1)$. Hence, $b_1(2j+2) - b_1(2j+1)$ equals the number of partitions $\lambda \in \mathcal{B}(2j+2)$ with $m_\lambda(1) = 0$. The transformation $\lambda \to \lambda/2$ is a bijection from the set of partitions $\lambda \in \mathcal{B}(2j+2)$ with $m_\lambda(1) = 0$ to $\mathcal{B}(j+1)$. This completes the proof. □

Corollary 4.4. *Let $n \geq 0$. Then*

$$\Delta b_1(2n) = 2(n+1)B(n+1).$$

Proof. From Proposition 4.3 we have

$$b_1(2n+2) - b_1(2n) = b_1(2n+2) - b_1(2n+1) = (2n+2)B(n+1).$$

□

In general, we have the following expression for $b_1(n)$.

Corollary 4.5. *For $n \geq 0$, $b_1(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} 2kB(k)$.*

Proof. Since $b_1(0) = 0$ and $b(2n+1) = b(2n)$ for all $n \geq 0$, iterating the statement of Corollary 4.4 gives the result. □

Proposition 4.6. *Let $n \geq 2$. Then $b_2(2n) \equiv 2 \pmod 4$.*

Proof. As in the proof of Corollary 4.2, we have

$$\sum_{\lambda \in \mathcal{B}(n)} m_\lambda(2) = \sum_{k=1}^{\lfloor n/2 \rfloor} B(n-2k)\,. \tag{6}$$

From Proposition 4.1(f), we have

$$\begin{aligned} b_2(2n) &= \sum_{\lambda \in \mathcal{B}(2n)} \left(\binom{2n - m_\lambda(1)}{2} + (n - m_\lambda(2)) \right) \\ &= \sum_{\lambda \in \mathcal{B}(2n)} \frac{4n^2 - 4nm_\lambda(1) + (m_\lambda(1))^2}{2} + \sum_{\lambda \in \mathcal{B}(2n)} \left(\frac{1}{2} m_\lambda(1) - m_\lambda(2) \right). \end{aligned}$$

Since $B(2k) = B(2k+1)$, it follows from (5) and (6) that the second sum above equals 0. Hence

$$\begin{aligned} b_2(2n) &= \sum_{\lambda \in \mathcal{B}(2n)} \frac{4n^2 - 4nm_\lambda(1) + (m_\lambda(1))^2}{2} \\ &= 2n^2 B(2n) - 4n \sum_{k=1}^{n} B(2n-2k) + \frac{1}{2} \sum_{\lambda \in \mathcal{B}(2n)} (m_\lambda(1))^2. \end{aligned}$$

We can show by induction that $B(j)$ is even for $j \geq 2$.

Next, we consider the last sum:

$$\sum_{\lambda \in \mathcal{B}(2n)} (m_\lambda(1))^2.$$

If $\lambda \in \mathcal{B}(2n)$, then $m_\lambda(1)$ is even. It is straightforward to see that the partitions λ with $m_\lambda(1) = 2k$ contribute $4k^2B(n-k)$ to the sum. Hence, since $B(0) = B(1) = 1$ we have

$$\frac{1}{2}\sum_{\lambda\in\mathcal{B}(2n)}(m_\lambda(1))^2 = \sum_{k=1}^{n} 2k^2B(n-k) \equiv 2n^2 + 2(n-1)^2 \equiv 2 \pmod 4.$$

Thus, $b_2(2n) \equiv 2 \pmod 4$. □

Remark 8. Using the interpretations above and the fact that $B(2n) = \sum_{k=0}^{n} B(k)$, we obtain

$$\begin{aligned} b_2(2n) &= 2n^2B(2n) - 4n\sum_{k=1}^{n} B(2n-2k) + \sum_{k=1}^{n} 2k^2B(n-k) \\ &= 2n^2\sum_{k=0}^{n} B(k) - 4n\sum_{k=1}^{n}\sum_{j=0}^{n-k} B(j) + \sum_{k=0}^{n-1} 2(n-k)^2B(k) \\ &= 2n^2\sum_{k=0}^{n} B(k) - 4n\sum_{k=0}^{n-1}(n-k)B(k) + \sum_{k=0}^{n-1} 2(n-k)^2B(k) \\ &= \sum_{k=0}^{n} 2k^2B(k). \end{aligned}$$

Comparing the expression for $b_2(2n)$ with that for $b_1(2n)$ given in Corollary 4.5, we see that $b_2(2n) > b_1(2n)$ for $n \geq 2$, and hence $b_2(n) > b_1(n)$ for $n \geq 3$.

We can also deduce that

$$\Delta b_2(n) = \begin{cases} 0 & \text{if } n \text{ even} \\ \frac{(n+1)^2}{2}B(\frac{n+1}{2}) & \text{if } n \text{ odd} \end{cases}$$

and

$$\Delta b_2(2n) = 2(n+1)^2B(n+1).$$

Next, we prove a recurrence relation for $\mathrm{num}_{\mathcal{B}}(2n, x)$.

Theorem 4.7. *For all $n \geq 1$,*

$$\mathrm{num}_{\mathcal{B}}(2n, x) = (1+x)^{2n} \cdot \mathrm{num}_{\mathcal{B}}(n, x^2) + f_{2n}(x) \cdot \mathrm{num}_{\mathcal{B}}(2n-2, x),$$

where

$$f_{2n}(x) = \prod_{i=1}^{\mathrm{val}_2(2n)} (1 + x^{2^i}).$$

Proof. Let

$$\mathcal{B}_0(2n) := \{\lambda \in \mathcal{B}(2n) : m_\lambda(1) = 0\}$$

and

$$\mathcal{B}_1(2n) := \{\lambda \in \mathcal{B}(2n) : m_\lambda(1) > 0\}.$$

Then

$$\mathcal{B}(2n) = \mathcal{B}_0(2n) \bigsqcup \mathcal{B}_1(2n).$$

The transformation $\lambda \to \lambda/2$ is a bijection from $\mathcal{B}_0(2n)$ to $\mathcal{B}(n)$. If $\lambda \in \mathcal{B}_1(2n)$, then $m_\lambda(1) \geq 2$. Thus, the transformation $\lambda \to \lambda \setminus (1^2)$ is a bijection from $\mathcal{B}_1(2n)$ to $\mathcal{B}(2n-2)$.

We define

$$\mathcal{H}_1(2n,x) = \sum_{\lambda\in\mathcal{B}_0(2n)} h_\lambda(x)\,,$$

and

$$\mathcal{H}_2(2n,x) = \sum_{\lambda\in\mathcal{B}_1(2n)} h_\lambda(x).$$

Then

$$\mathrm{num}_{\mathcal{B}}(2n,x) = \mathcal{H}_1(2n,x) + \mathcal{H}_2(2n,x).$$

If $\lambda \in \mathcal{B}_0(2n)$, then $\lfloor \frac{2n}{2^i} \rfloor - m_\lambda(2^i) = \lfloor \frac{n}{2^{i-1}} \rfloor - m_{\lambda/2}(2^{i-1})$ for all $i \geq 1$, and thus

$$h_\lambda(x) = h_{\lambda/2}(x^2)\cdot(1+x)^{2n}\,.$$

Hence, $\mathcal{H}_1(2n,x) = (1+x)^{2n}\cdot \mathrm{num}_{\mathcal{B}}(n,x^2)$.

If $\lambda \in \mathcal{B}_1(2n)$, then

$$\begin{aligned} h_\lambda(x) =& \prod_{i\geq 0}(1+x^{2^i})^{\lfloor 2n/2^i\rfloor - m_\lambda(2^i)} \\ =& \prod_{i\geq 1}(1+x^{2^i})^{\lfloor 2n/2^i\rfloor - \lfloor (2n-2)/2^i\rfloor}\cdot h_{\lambda\setminus(1^2)}(x) \\ =& f_{2n}(x)\cdot h_{\lambda\setminus(1^2)}(x)\,. \end{aligned}$$

Hence, $\mathcal{H}_2(2n,x) = f_{2n}(x)\cdot \mathrm{num}_{\mathcal{B}}(2n-2,x)$. □

Remark 9. Recall that $d(n) = \deg \mathrm{num}_{\mathcal{B}}(n,x) = \deg h_\lambda(x)$ for all $\lambda \in \mathcal{B}(n)$. Clearly

$$\deg f_{2n}(x) = 2^{\mathrm{val}_2(2n)} + \ldots + 2 = 2^{\mathrm{val}_2(2n)+1} - 2\,,$$

and

$$d(2n) = \deg \mathcal{H}_1(2n,x) = \deg \mathcal{H}_2(2n,x)\,.$$

Next, we deduce recurrence relations and a formula for $d(n)$.

Proposition 4.8. *The following hold.*

(a) For all even $n \geq 2$,

$$d(n) - d(n-2) = 2^{\mathrm{val}_2(n)+1} - 2.$$

(b) For all $n \geq 2$,

$$d(n) = 2\left(d(\lfloor \frac{n}{2} \rfloor) + \lfloor \frac{n}{2} \rfloor\right).$$

(c) If $n = \sum_i a_i 2^i$ is the binary representation of n, then

$$d(n) = \sum_i i\cdot a_i 2^i.$$

Remark 10. The recurrence (b) computes $d(n)$ in $\log_2(n)$ steps. The expressions in (b) and (c) are given without proof in [5, A136013] (for the sequence $d(n)/2$).

Proof. (a) For $n = 2m \geq 2$,

$$\begin{aligned} d(n) = \deg \mathcal{H}_2(2m,x) &= \deg f_{2m}(x) + \deg \mathrm{num}_{\mathcal{B}}(2m-2,x) \\ &= 2^{\mathrm{val}_2(n)+1} - 2 - d(n-2)\,. \end{aligned}$$

(b) First assume $n = 2m$ to be even. Then

$$d(n) = \deg \mathcal{H}_1(2m,x) = \deg(1+x)^{2m} + \deg \mathrm{num}_{\mathcal{B}}(m,x^2)$$

$$= 2m + 2d(m) = 2\left(d(\frac{n}{2}) + \frac{n}{2}\right) .$$

For n odd, $d(n) = d(n-1)$ and $\lfloor \frac{n}{2} \rfloor = \lfloor \frac{n-1}{2} \rfloor$, which completes the proof.

(c) This follows by induction using (b). □

Next, we give some evidence for the main conjecture of this section, which we rephrase below.

Conjecture 7. *For all $n \geq 2$ and for all $2^i \leq n$ we have*

$$(1 + x^{2^i}) \nmid \mathrm{num}_{\mathcal{B}}(n, x).$$

Proposition 4.9. *Conjecture 5 is equivalent to Conjecture 7.*

Proof. By Lemma 2.1, the polynomials $(1+x^{2^i})$ are irreducible for all $i \geq 0$. Thus, $\mathrm{den}_{\mathcal{B}}(n, x) = \prod_{i\geq 0}(1+x^{2^i})^{\lfloor n/2^i \rfloor}$ is the decomposition of $\mathrm{den}_{\mathcal{B}}(n, x)$ into irreducible factors. Hence, any irreducible factor of $\gcd(\mathrm{den}_{\mathcal{B}}(n, x), \mathrm{num}_{\mathcal{B}}(n, x))$ must be of the form $(1 + x^{2^i})$ for $2^i \leq n$. □

Theorem 4.10. *Conjecture 7 holds for $n = 2^k$ and $n = 2^k + 1$ for $k \geq 0$.*

Proof. If $n = 2^k$, and $i \leq k$ then $\lfloor n/2^i \rfloor = 2^{k-i}$. Let $\lambda \in \mathcal{B}(2^k)$ be the partition with 2^{k-i} parts all equal to 2^i. Then $(1 + x^{2^i}) \nmid h_\lambda(x)$. However, $(1 + x^{2^i}) \mid h_\mu(x)$ for all $\mu \neq \lambda$. Hence, $(1 + x^{2^i}) \nmid \mathrm{num}_{\mathcal{B}}(2^k, x)$. The statement for $n = 2^k + 1$ follows from Proposition 4.1 (a). □

As in Lemma 3.14, $(1 + x^{2^i}) \mid \mathrm{num}_{\mathcal{B}}(n, x)$ if and only if $(1 + x^{2^i}) \mid \mathrm{num}_{\mathcal{B}}(r, x)$, where $r = n - 2^i\lfloor \frac{n}{2^i} \rfloor$. Then, one can show that for $n \geq 2$, $(1 + x^{2^i}) \nmid \mathrm{num}_{\mathcal{B}}(n, x)$ for $i = 0, 1$ and 2.

5. Sums of powers of subsum polynomials

Let $\beta(n, x, m) = \sum_{\lambda\vdash n} \mathrm{sp}(\lambda, x)^m$. So far we have studied $\beta(n, x, -1) = \mathrm{sr}(n, x)$.

Let $P_k(n)$ be the number of partitions of n into k sorts of parts; the order does not matter for parts of different sizes, but it does matter for parts of the same size and of different sorts. For example, for $n = 3$ and $k = 2$, the 14 partitions are

$$(3_a), (3_b), (2_a, 1_a), (2_a, 1_b), (2_b, 1_a), (2_b, 1_b),$$
$$(1_a, 1_a, 1_a), (1_b, 1_a, 1_a), (1_a, 1_b, 1_a), (1_a, 1_a, 1_b), (1_a, 1_b, 1_b),$$
$$(1_b, 1_a, 1_b), (1_b, 1_b, 1_a), (1_b, 1_b, 1_b).$$

(See A246935 in [5].)

Proposition 5.1. *For $n \geq 1$,*

(a) $\beta(n, 1, m) = P_{2^m}(n)$,

(b) $\beta(2n, -1, m) = P_{2^m}(n)$ and $\beta(2n-1, -1, m) = 0$.

Proof. (a) First note that $\mathrm{sp}(\lambda, 1) = \prod_{i=1}^{n} 2^{m_\lambda(i)} = 2^{\sum m_\lambda(i)} = 2^{\ell(\lambda)}$, where $\ell(\lambda)$ is the number of parts of λ. Hence,

$$\beta(n, 1, m) = \sum_{\lambda\vdash n} 2^{m\ell(\lambda)}.$$

To see that this equals $P_{2^m}(n)$, notice that from each ordinary partition λ of n (that is, a partition with parts of a single sort), we can create $2^{m\ell(\lambda)}$ partitions with 2^m

sorts of parts. There are 2^m choices for the sorts for the first part, 2^m choices for the sorts for the second part, etc.

(b) If λ has an odd part, then $\mathrm{sp}(\lambda, -1) = 0$. If $\mathcal{P}_e(n)$ is the set of partitions of n with all parts even, then

$$\beta(n, -1, m) = \sum_{\lambda \in \mathcal{P}_e(n)} \mathrm{sp}(\lambda, -1)^m.$$

Since all partitions of $2n-1$ have at least one odd part, $\beta(2n-1, -1, m) = 0$.

The transformation $\lambda \to \lambda/2$ is a bijection from $\mathcal{P}_e(2n)$ to the set of all partitions of n. If $\lambda \in \mathcal{P}_e(2n)$, we have $\mathrm{sp}(\lambda, -1) = \mathrm{sp}(\lambda, 1) = \mathrm{sp}(\lambda/2, 1)$. Thus $\beta(2n, -1, m) = \beta(n, 1, m)$ and the result follows from (a). □

Proposition 5.2. *Let $n \geq 1$ and denote by $\beta'(n, x, m)$ the derivative of $\beta(n, x, m)$ with respect to x. Then*

$$\mathrm{val}_2(\beta'(n, 1, k)) = \mathrm{val}_2(n) + \mathrm{val}_2(k) + k - 1.$$

Proof. We have

$$\beta'(n, x, k) = \sum_{\lambda \vdash n} k \cdot \mathrm{sp}(\lambda, x)^{k-1} \mathrm{sp}'(\lambda, x).$$

Using logarithmic differentiation, we have

$$\mathrm{sp}'(\lambda, x) = \mathrm{sp}(\lambda, x) \sum_{i=1}^{n} m_\lambda(i) \frac{i x^{i-1}}{1 + x^i}.$$

Evaluating at $x = 1$, we obtain

$$\beta'(n, 1, k) = \sum_{\lambda \vdash n} k \cdot \mathrm{sp}(\lambda, 1)^k \sum_{i=1}^{n} m_\lambda(i) \frac{i}{2} = \frac{1}{2} nk \sum_{\lambda \vdash n} 2^{k\ell(\lambda)} = \frac{1}{2} nk 2^k \sum_{\lambda \vdash n} 2^{k(\ell(\lambda)-1)}.$$

Since $\lambda = (n)$ is the only partition of n with $\ell(\lambda) = 1$, it follows that $\sum_{\lambda \vdash n} 2^{k(\ell(\lambda)-1)}$ is odd. Hence, $\mathrm{val}_2(\beta'(n, 1, k)) = \mathrm{val}_2(n) + \mathrm{val}_2(k) + k - 1$. □

6. Open Questions

Analogously to $\mathrm{sr}_{\mathcal{B}}(n, x)$ for the binary partitions of n, we can define $sr_{\mathcal{A}}(n, x)$ for any subset $\mathcal{A}(n)$ of partitions of n and study the questions posed in this article. We offer the following conjectures.

Let $\mathcal{O}(n)$ be the set of odd partitions of n, that is, partitions whose parts are all odd.

Conjecture 8. *For $n \geq 1$, $\mathrm{num}_{\mathcal{O}}(n, -1) = \mathbf{o}(n!)$.*

Let $\mathcal{T}(n)$ be the set of ternary partitions of n, that is, partitions whose parts are powers of 3.

Conjecture 9. *Let $s(n) = \mathrm{num}_{\mathcal{T}}(n, -1)$. Then $s(1) = s(2) = 1$ and $s(3n) = s(3n+1) = s(3n+2) = 3^{\mathrm{val}_3((3n)!)}$.*

Conjecture 10. *Let $t(n) = \mathrm{num}_{\mathcal{T}}(n, 1)$. Then $t(1) = t(2) = 1$ and $t(3n) = t(3n+1) = t(3n+2)$. Also, $t(n) = \Delta(t(3n-2)/2^{2n})$.*

Finally, we note that one can replace x^i in $\mathrm{sp}(\lambda, x)$ and $\mathrm{sr}(n, x)$ by $x^{\overline{i}}$ or $x^{\underline{i}}$, where $\overline{i}$ and $\underline{i}$, are the rising and falling factorials, respectively. Conjecturally, these polynomials have interesting properties, and we hope the interested reader will explore them.

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